

Introduction to ER Diagrams

An **Entity-Relationship Diagram (ER Diagram or ERD)** is a graphical representation used to design and visualize the structure of a database. It helps database designers understand how data is organized, how different entities interact, and how information flows within a system.

ER diagrams are widely used during the **conceptual design phase** of database development because they provide a simple and intuitive way to model real-world scenarios.

Why ER Diagrams are Important

ER diagrams provide several advantages in database design:

- They offer a visual representation of data and relationships.
 - They simplify communication between developers, analysts, and clients.
 - They help identify missing data requirements.
 - They serve as a blueprint for creating relational database tables.
 - They reduce redundancy and improve data consistency.
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ER Model in Database Design Process

Database design generally follows these steps:

Step 1: Requirement Gathering

Understand what information needs to be stored and how users will interact with the system.

Step 2: Conceptual Design

Create an ER diagram representing entities, attributes, and relationships.

Step 3: Logical Design

Convert the ER diagram into relational schemas.

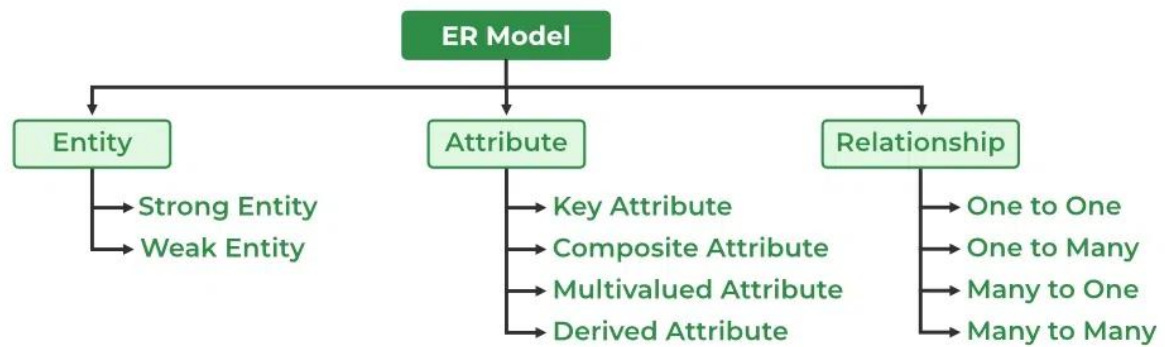
Step 4: Physical Design

Define storage structures, indexes, constraints, and optimization strategies.

Components of an ER Diagram

An ER Diagram consists of three major components:

1. Entities
2. Attributes
3. Relationships



1. Entity

An **Entity** is a real-world object, person, place, event, or concept about which information is stored in a database.

Entities act as building blocks of a database.

Examples

Domain	Entity Examples
Education	Student, Teacher, Course
Banking	Customer, Account
E-Commerce	Product, Customer, Order
Hospital	Doctor, Patient

Representation

Entities are represented by a **Rectangle**.

```
+-----+
| Student |
+-----+
```

Types of Entities

A) Strong Entity

A strong entity can be uniquely identified using its own attributes.

It possesses a primary key.

Characteristics

- Independent entity
- Has its own primary key
- Represented using a single rectangle

Example

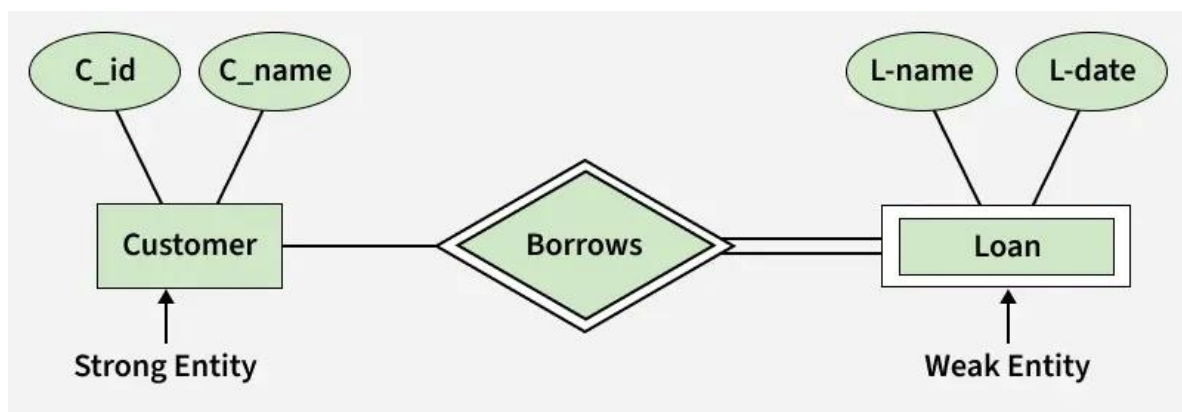
Student

Attributes:

- StudentID
- Name
- Email

```
+-----+
| Student |
+-----+
```

B) Weak Entity



A weak entity cannot exist without another entity.

It does not possess sufficient attributes to uniquely identify itself.

It depends upon a strong entity.

Characteristics

- Dependent entity
- No complete primary key
- Represented using double rectangles

Example

Employee → Dependent

A dependent cannot exist without an employee.

```
+=====+
| | Dependent | |
+=====+
```

Entity Set

An **Entity Set** is a collection of similar entities.

Example:

Student Entity Set

Student1

Student2

Student3

Student4

ER diagrams represent entity sets rather than individual records.

2. Attributes

Attributes describe the properties of entities.

For a Student entity:

- StudentID
- Name
- Address
- DateOfBirth
- MobileNo

Attributes are represented by **Ovals**.

(Name)

```
      |
+-----+
| Student |
+-----+
```

Types of Attributes

1. Key Attribute

A key attribute uniquely identifies an entity.

Example

StudentID

Representation

Underlined Oval

(StudentID)

2. Composite Attribute

Composite attributes can be divided into smaller subparts.

Example

Address

Contains:

- Street
- City
- State
- Country

(Address)
/ | \
(Street)(City)(State)

3. Multivalued Attribute

Stores multiple values for one entity.

Example

Phone Numbers

A student may have:

- Personal Number
- Home Number

Representation:

Double Oval

((PhoneNo))

4. Derived Attribute

Can be calculated from existing attributes.

Example

Age

Derived from Date of Birth.

Representation:

Dashed Oval

--- Age ---

Complete Student Entity Example

(Name)

|

(DOB) Student (Email)

|

(__StudentID__)

|

((PhoneNo))

3. Relationship

Relationships describe associations between entities.

Example

A student enrolls in a course.

Entities:

Student

Course

Relationship:

Enrolls

Representation

Relationships are represented using Diamonds.

```
+-----+   ◆ Enrolls ◆   +-----+
| Student |----->| Course |
+-----+           +-----+
```

Relationship Set

A relationship set contains all relationships of the same type.

Examples

Student S1 enrolled in Course C1

Student S2 enrolled in Course C2

Student S3 enrolled in Course C3

Degree of Relationship

Degree refers to the number of participating entity sets.

Unary Relationship

Only one entity participates.

Example

Person Marries Person

Person

|

Married

|

Person

Binary Relationship

Two entities participate.

Example

Student Enrolls Course

Student ---- Enrolls ---- Course

Ternary Relationship

Three entities participate.

Example

Doctor Prescribes Medicine to Patient

Doctor

\

Prescribes

/ \

Patient Medicine

N-ary Relationship

More than three entities participate.

Examples:

- Supplier
 - Product
 - Warehouse
 - Delivery Agent
-

Cardinality Constraints

Cardinality defines how many instances of one entity can be associated with instances of another entity.

1. One-to-One (1:1)

Each entity is associated with exactly one entity.

Example

Person ↔ Passport

Person ----- Passport

1 : 1

2. One-to-Many (1:M)

One entity relates to many entities.

Example

Department → Employees

Department ----- Employee

1 : M

3. Many-to-One (M:1)

Many entities relate to one entity.

Example

Many Students belong to one Department.

Student ----- Department

M : 1

4. Many-to-Many (M:N)

Multiple entities relate to multiple entities.

Example

Students and Courses

Student ----- Course

M : N

One student can take many courses.

One course can have many students.

Participation Constraints

Participation constraints specify whether participation in a relationship is mandatory.

Total Participation

Every entity must participate.

Represented by **double lines**.

Example

Every student must enroll in at least one course.

Student **=====** Enrolls **=====** Course

Partial Participation

Participation is optional.

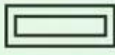
Represented by **single lines**.

Example

Some courses may not have students enrolled.

Student — Enrolls — Course

Symbols Used in ER Diagrams

Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Symbol

Rectangle

Double Rectangle

Oval

Underlined Oval

Double Oval

Dashed Oval

Diamond

Meaning

Entity

Weak Entity

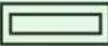
Attribute

Key
Attribute

Multivalued
Attribute

Derived
Attribute

Relationship

Figures	Symbols	Represents
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Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Symbol

Meaning

Double Diamond

Identifying Relationship

Single Line

Partial Participation

Double Line

Total Participation

Steps to Draw an ER Diagram

Step 1: Identify Entities

Example:

- Customer
- Product
- Order

Step 2: Identify Relationships

Examples:

- Customer places Order
- Order contains Product

Step 3: Add Attributes

Customer

- CustomerID
- Name
- Email

Product

- ProductID
- Price
- Category

Step 4: Define Primary Keys

Underline the unique identifier.

Example:

- CustomerID
- ProductID
- OrderID

Step 5: Define Cardinalities

Examples:

- One Customer can place many Orders.
- One Order can contain many Products.

Step 6: Review and Remove Redundancies

Ensure there are:

- No duplicate entities
- No unnecessary relationships
- Proper naming conventions

Example: ER Diagram for an E-Commerce System

Entities

Customer

Product

Order

Payment

Category

Relationships

Customer **places** Order

Order **contains** Product

Order **has** Payment

Product **belongs to** Category

Cardinalities

Relationship	Cardinality
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Customer → Order	1:M
------------------	-----

Order → Payment	1:1
-----------------	-----

Order → Product	M:N
-----------------	-----

Category → Product	1:M
--------------------	-----

Summary

ER diagrams are an essential tool in database design because they:

- Provide a clear visualization of database structure.
- Help convert real-world requirements into database schemas.
- Define entities, attributes, relationships, and constraints effectively.
- Simplify the transition from conceptual design to implementation.
- Improve database consistency, scalability, and maintainability.

These concepts form the foundation for designing efficient databases for applications such as **E-Commerce Systems, Hospital Management Systems, Banking Systems, Library Management Systems, and Student Information Systems.**